CAD 4.0

TRACK - CYBERSECURITY

TEAM NAME : PHISH POLICE

PRODUCT: FRAUD SHIELD AI

TEAM MEMBERS:

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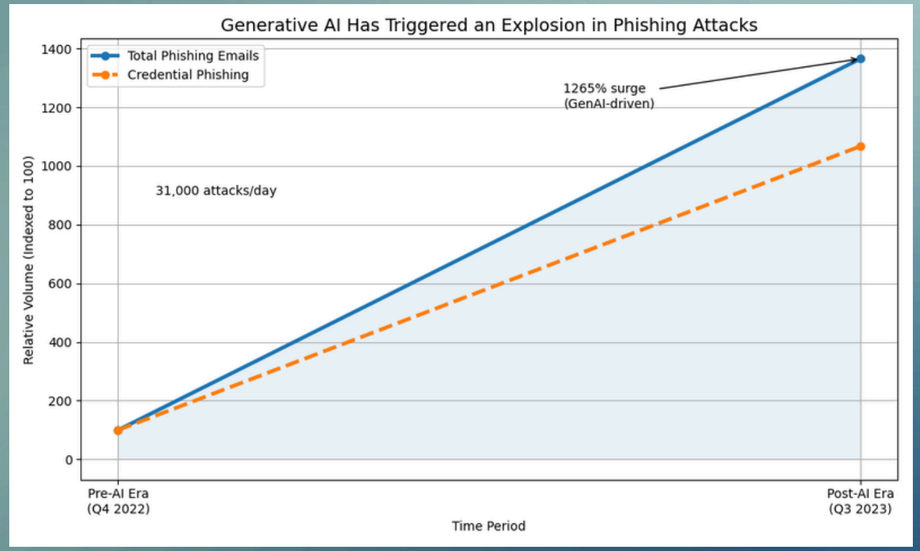
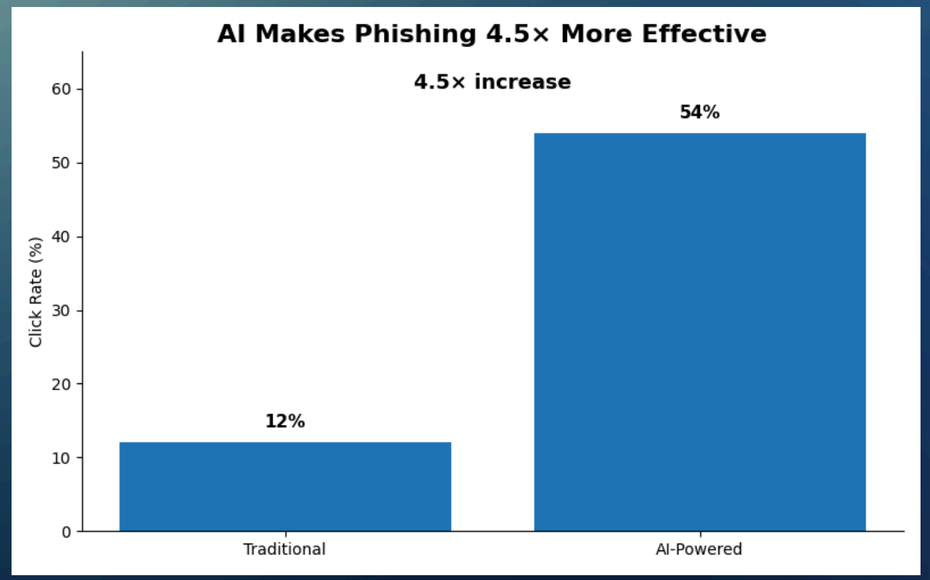
PROBLEM STATEMENT-

Scammers have upgraded to Sovereign AI Agents that create unique, perfectly, written traps for every victim in real-time. Legacy security is failing because it relies on 'Known Bad Lists' to fight Autonomous Attacks that have never existed before. We are no longer fighting simple scripts; we are facing an Adversarial Intelligence that outthinks the user at the point of attack

- AI scrapes social data to build a bespoke psychological profile for every victim. **Result: 54% Click Rate.**
- Voice clones with **95% emotional accuracy** bypass verbal trust and KYC verification.
- Malicious links that **"detect" sandboxes** and act normal until they reach the human target



SOURCE: SLASHNEXT (2023)



MICROSOFT DIGITAL DEFENSE REPORT (2025)

PROJECTED ₹70,000 CRORE LOSS IN INDIA THIS YEAR TO DEEPFAKE FRAUD ALONE.

PROPOSED SOLUTION

One input. One threat score. One clear decision.

One AI shield that detects scams across text, email, URLs, voice calls, video deepfakes, and file attachments in real time.

LIVE MODULE

Text Analysis

Detects phishing intent, urgency, and AI-generated scam language patterns

LIVE MODULE

Credential Detection

Flags requests for OTPs, passwords, Aadhaar, PAN, and bank details

BEHAVIORAL

URL Sandbox

Checks suspicious links using SSL, WHOIS, and safe browser sandboxing

THREAT OUTPUT

Threat Score

Unified verdict from all active detectors

92 HIGH RISK SCAM

Credential theft and impersonation detected

FILE GUARD

File & Attachment Scan

Scans files with YARA, ClamAV, and threat intelligence feeds

REAL-TIME

Voice Fraud Detection

Analyzes live call transcripts, acoustic signals, and deepfake indicators

CORE DEFENSE

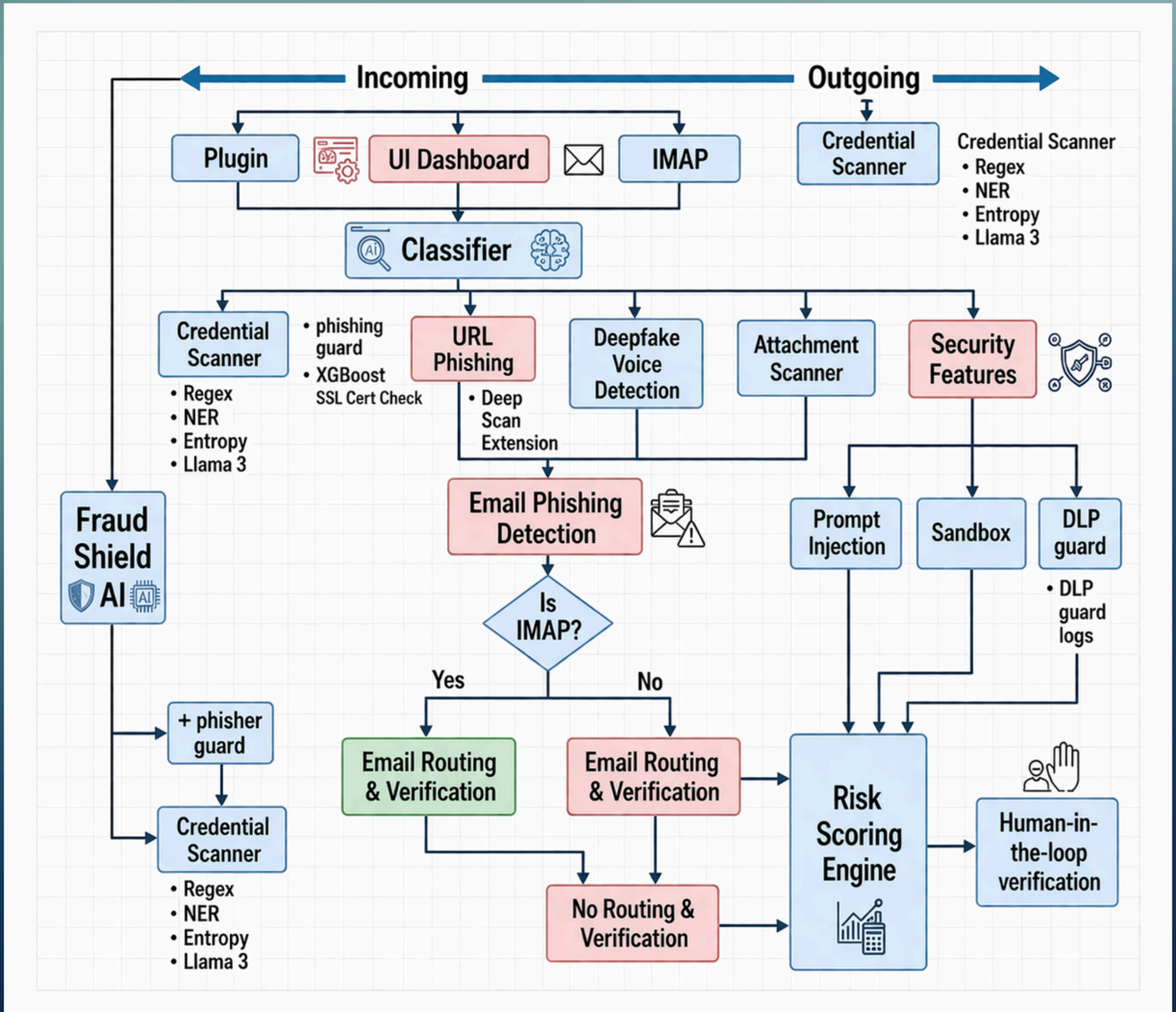
Self-Protecting AI

Shadow Guard and DLP Guard block prompt injection and data leaks

ARCHITECTURAL WORKFLOW DIAGRAM

Components

- Multi-source input (Plugin, UI, IMAP)
- Central classification engine
- Credential detection (Regex, NER, Entropy, LLM)
- URL phishing detection (XGBoot, SSL, Deep Scan)
- Deepfake voice detection
- Attachment scanning
- Prompt injection & DLP protection
- Unified phishing detection
- Smart AI Generated Email Detection
- Fidelity Ranking
- Human-in-the-loop feedback



TECH STACK

MACHINE LEARNING

- python 3.13
- XGBoost: Structured URL classification and phishing heuristics.
- ChromaDB: Vector Database for storing and matching incoming threats against a massive library of known scam patterns.
- OpenAI Whisper (Faster-Whisper): Real-time Speech-to-Text (STT) for processing live voice calls.
- Librosa + SpeechBrain: Detect synthetic "artifacts" in Deepfake Voice Detection.

ACTION-FOCUSED (BEST FOR BULLETS)

- URL Sandbox: Playwright (Real-time Detonation)
- Credential Scanner: Microsoft Presidio (PII/NER)
- File Security: YARA & ClamAV (Malware Signatures)
- Live Integration: IMAP (Real-time Email Hooks)
- Shadow Guard: NeMo Guardrails (Anti-Injection)

CONTAINERIZATION & INFRASTRUCTURE

- Docker: Creating the sandbox enviroment
- PostgreSQL: Storing threat logs and human-in-the-loop verification data.
- Ollama: High-speed inference engines

WEB DEVELOPEMENT

- Next.js 14 / React (frontend)
- Tailwind CSS + Shadcn/UI (stylesheet)
- FastAPI (Python): (backend)
- WebSockets: push live Risk Scores and "Threat Alerts" to the dashboard instantly.

What Makes Us Different

Feature	Truecaller	Gmail	Antivirus	Zora AI	Fraud Shield AI Highlighted advantage
AI-Generated Text Detection	✗	✗	✗	✗	✓
Real-Time Voice Fraud Detection	Caller ID only	✗	✗	✓	✓
Deepfake-Aware Protection	✗	✗	✗	Voice only	Voice + Video Roadmap
Prompt Injection Defense	✗	✗	✗	✓	✓
DLP / Output Leak Protection	✗	✗	✗	✗	✓
Unified Multi-Channel Coverage	Partial	Email only	Files only	Strong	Text, Email, URL, Voice, File
Human Feedback Learning Loop	Report spam	Report spam	✗	✗	✓
Threat Score With Reasoning	✗	Basic	Signature-based	Partial	Full score + reasoning

Fraud Shield AI combines real-time detection, self-protection, and continuous learning in one system.

IMPACT AND FEASIBILITY

- **Sub-200ms Latency:** Real-time interception optimized for live calls and high-speed mobile networks.
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- **Scalable Hybrid AI:** XGBoost and Llama 3 deliver elite accuracy with minimal compute costs.
- **Secure "Detonation":** Docker-isolated sandboxing safely analyzes malicious links without user risk.
- **Rapid Integration:** Modular API-first design (pip install) for deployment in banking apps in days.
- **Preventative ROI:** Shifting from "tracking losses" to saving ₹70,000 Cr at the point of attack.
- **Video deepfake detection:** Future module for facial landmark and lipsync based deepsync analysis

THANK YOU